

SAI VISHNU VAMSI SENAGASETTY

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Software Engineer with 5 years of experience building AI/ML systems, full-stack applications, and end-to-end RAG chatbots, seeking full-time roles in applied machine learning or AI product development.

EDUCATION

Master of Science in Computer Science, University of Houston August 2023 - May 2025
Relevant Coursework: Machine Learning, Deep Learning, Big Data Analytics, Cloud Computing, AI Systems

Bachelor of Technology in Computer Science, SRM University, AP June 2019 – May 2023

PROJECT EXPERIENCE

RAG Chatbot: Personal Portfolio Assistant using LangChain July 2025 - August 2025

- Developed a fully functional Retrieval-Augmented Generation (RAG) chatbot powered by LangChain and Together.ai API, serving personalized responses from a structured knowledge base.
- Engineered custom embeddings and vector similarity search to retrieve relevant document chunks with high accuracy across personal data.
- Integrated Replit-hosted frontend with Render-deployed Flask backend, enabling low-latency LLM interaction with live endpoints.
- Achieved <1s average response time and 98% accuracy on portfolio Q&A queries; built for scalable deployment in static portfolio sites.

Benchmarking Model Deployment: TorchServe vs TensorFlow Serving March 2025 - April 2025

- Designed and deployed CNN and RNN models using TorchServe and TensorFlow Serving to benchmark latency and throughput under simulated loads.
- Built a REST API load simulator using multithreading to model real-world request flows (5–50 req/sec).
- Used Azure ML SDK v2 to configure environments, deploy endpoints, and manage traffic.
- Achieved 11.6% lower latency with TorchServe (avg. 0.042s) over TensorFlow (avg. 0.047s) under medium load; throughput reached 45 req/sec with 100% success rate.
- Gained experience with VM quotas, endpoint versioning, and model compatibility on cloud.

WORK EXPERIENCE

AI/ML Research Intern August 2024 – December 2024

ML Engineer January 2025 – Present

- Proposed AI integration strategies to align product roadmap with mobility innovation goals.
- Partnered with cross-functional teams to accelerate timelines by 10% through automation and ML-driven feature planning.
- Contributed to the development of the core Flutter-based app, improving cross-platform performance and reducing UI/UX issues by 25%.
- Engineered an ML-based voice-to-ride-booking pipeline using speech-to-text transcription and NLP, enabling hands-free ride scheduling in the driver app.

Instructional Assistant, University of Houston September 2024 – Present

- Modernized university websites with HTML/CSS and AI-assisted workflows, improving content update efficiency by 75%.
- Built bio pages, galleries, and multimedia sections using hand-coded solutions, removing reliance on third-party plugins.

- Created flyers and visuals with AI-enhanced Adobe tools, streamlining design workflows and accelerating delivery by 25%.
- Created customized visual themes and interactive components tailored to department needs, enhancing user engagement.
- Introduced AI capabilities to cross-functional teams, accelerating workflows and improving collaboration across communications and marketing.

NLP Research Assistant

University of Houston

January 2024 – July 2024

Houston, TX

- Investigated rising misinformation in “pink slime” journalism by analyzing 600k+ news article sentences across multiple categories to detect deceptive patterns in textual style and structure.
- Built an NLP pipeline using Python, NLTK, SpaCy, and TextBlob to extract stylistic features, applied TF-IDF vectorization and dimensionality reduction.
- Produced a clean, labeled dataset of 27K sentences, uncovered up to 8 distinct clusters in categories like Politics and Business. Pink slime articles exhibited 3× sentiment exaggeration, lower linguistic diversity.
- Discovered that pink slime journalism favors overly positive tone, minimal sentence depth, and active voice dominance. Visualizations (t-SNE, heatmaps, boxplots) revealed stark readability and stylistic contrasts — laying the groundwork for misinformation detection tools and future classification models.

PUBLICATIONS

- [Nighttime Object Detection: A Night-Patrolling Mechanism Using Deep Learning](#) March 2023
- [State of Health of Lithium-ion Batteries by Data-driven Technique with Optimized Gaussian Process Regression](#) April 2023

SKILLS

Programming	Python, R, SQL, C++, Bash, HTML/CSS, JavaScript, Flutter, CUDA
Analytics / ML	Machine Learning, Predictive Analytics, Time-Series Modeling, Statistics, Data Visualization, Text Processing, Topic Modeling, LLM Integration, Model Tuning, Prompt Engineering, GPU Computing, CI/CD, Workflow Orchestration, ETL Pipelines, Web Scraping, Model Deployment
AI/ML Frameworks	TensorFlow, PyTorch, Scikit-learn, ONNX, LangChain, RAG, Sentence Transformers, FAISS
Tools	Apache Airflow, Docker, Docker Compose, Kubernetes, Git, GitHub, GitHub Actions, Databricks, Spark, Hadoop, Azure ML, Palantir Foundry, Groq API, Power BI, Tableau, Excel, CMS, Jupyter, Google Colab, VS Code, Linux
Soft Skills	Agile/Scrum, Problem Solving, Communication, Team Collaboration, Documentation
Certifications	Oracle Cloud Infrastructure 2025 Certified Generative AI Professional , Machine Learning A-Z™ (Python & R) , Excel Skills for Business (JPMC) , Analyzing Marketing Campaigns with Pandas , Data Manipulation in SQL

EXTRA-CURRICULAR ACTIVITIES

- Graduate Mentor, SACNAS-UH August 2024 - May 2025
Provided 1:1 mentorship to 10+ undergraduates in research and academics.
- Student Council Associate, SRM University February 2022 - August 2022
Contributed to 5+ campus initiatives focused on student engagement and infrastructure.
- Member, Gaming and Welfare Clubs, SRM University August 2021 - May 2023
Organized 8+ events, boosting student engagement by 30%.